BASIC ELECTRONICS ENGINEERING • PART B

UNIT I

Semiconductor Devices

From vacuum tubes to nano-electronics, diodes, Zener diode, BJT and the CE amplifier

WHAT YOU WILL LEARN

1 Evolution of electronics
2 PN junction diode & its characteristics
3 Zener effect & Zener diode
4 Bipolar Junction Transistor (BJT)
5 CB, CE, CC configurations
6 Input / output characteristics
7 Small-signal CE amplifier
8 Formulas, summary & exam questions

1 Evolution of Electronics: Vacuum Tubes to Nano-electronics

Electronics is the branch of engineering that controls the flow of electrons through vacuum, gases and (mostly today) **semiconductors** to process information and power. Every phone, TV and computer is a descendant of one idea: *use a small signal to control a large flow of electrons*.

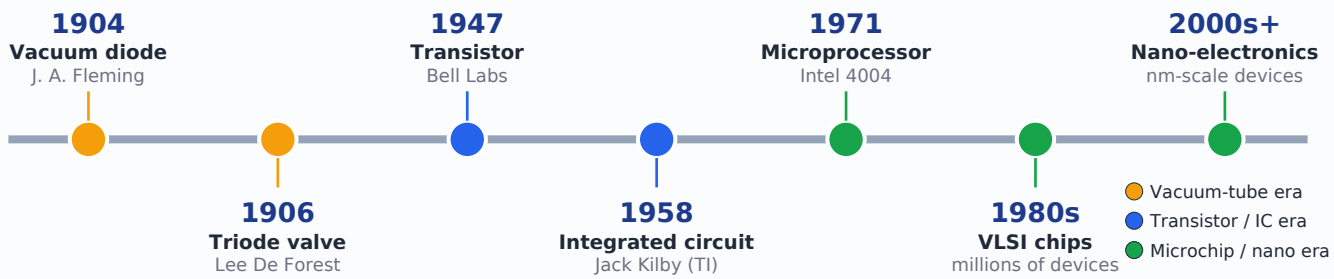


Fig 1.1 Milestones in the evolution of electronics

Vacuum tube: the first electronic device

- A heated **cathode** releases electrons (*thermionic emission*); a positive **anode (plate)** collects them. A **grid** in a triode controls the current, hence it amplifies.
- Used in early radio, TV, radar and the first computers (ENIAC, ~18,000 tubes).

Property	Vacuum tube	Transistor / IC
Size	Large and bulky	Tiny (nm to mm)
Power	High (heater needed)	Very low
Warm-up time	Needs seconds to heat	Instant on
Life and strength	Short life, fragile glass	Very long, rugged
Operating voltage	Hundreds of volts	A few volts
Cost (mass production)	High	Extremely low

Integration levels and nano-electronics

Era	Approx. transistors per chip	Example
SSI / MSI	up to a few hundred	Logic gates, counters
LSI	thousands to ~100,000	Early microprocessors, memories
VLSI / ULSI	lakhs to billions	Modern CPUs, mobile SoCs

- **Nano-electronics** deals with devices whose key dimensions are in the nanometre range (1–100 nm), where **quantum effects** become important.

- Examples: **FinFET / gate-all-around** transistors, **carbon-nanotube** and **graphene** devices, quantum dots, nanowires, memristors.
- Benefits: higher speed, lower power, huge density. Challenges: leakage, heat, tunnelling and fabrication cost.

Applications of nano-electronics
Smartphone processors, AI accelerators, flash memory, medical biosensors, flexible displays, solar cells.

Did you know?
- The ENIAC computer (1946) used about 18,000 vacuum tubes and filled a large hall; a modern phone chip packs **billions** of transistors in a fingernail-sized die.
- A human hair is ~80,000 nm wide, while today's smallest transistor features are only a few nanometres.

2 PN Junction Diode

A **semiconductor** (Si, Ge) conducts better than an insulator but worse than a conductor. Adding impurities (**doping**) makes it *extrinsic*.

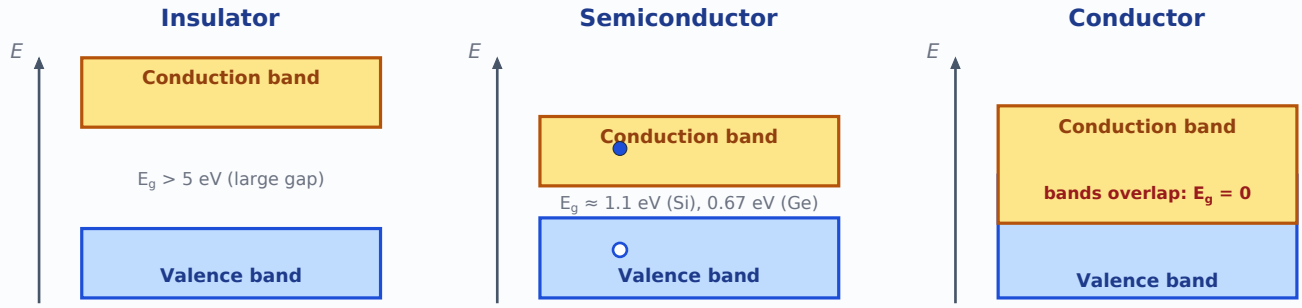


Fig 1.2 Energy-band view: the gap between valence and conduction bands decides how well a material conducts

	N-type	P-type
Dopant	Pentavalent (P, As, Sb): *donor*	Trivalent (B, Al, Ga): *acceptor*
Majority carriers	Free electrons	Holes
Minority carriers	Holes	Electrons
Charge of crystal	Neutral overall	Neutral overall

DEFINITION
A **PN junction diode** is formed by joining P-type and N-type semiconductors. It is a **two-terminal, one-way device**: Anode (P) and Cathode (N). Symbol: ▶| (current flows easily from anode to cathode).

Formation of the depletion region

- Electrons from N-side diffuse across to the P-side; holes diffuse the other way and they **recombine** near the junction.
- This leaves uncovered fixed ions, which form a thin **depletion region** with no free carriers.
- The ions create an internal electric field, the **barrier potential**: about **0.7 V for Si** and **0.3 V for Ge** (at 25 °C).

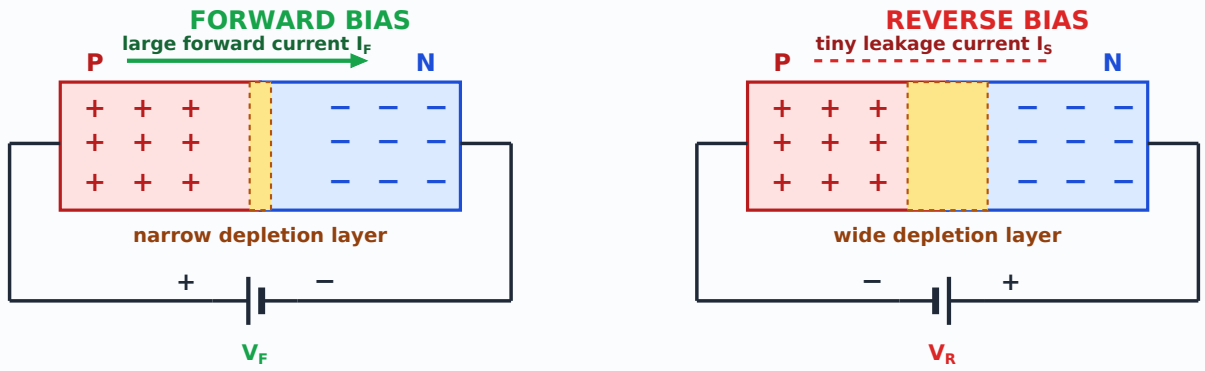


Fig 1.3 PN junction under forward bias (left) and reverse bias (right)

	Forward bias (P to +, N to −)	Reverse bias (P to −, N to +)
Depletion region	Becomes **narrow**	Becomes **wide**
Barrier potential	Reduced	Increased
Current	Large; flows once V > knee voltage	Very small leakage (minority carriers) I_S
Diode acts as	Closed switch (ON)	Open switch (OFF)

3 V-I Characteristics of a PN Junction Diode

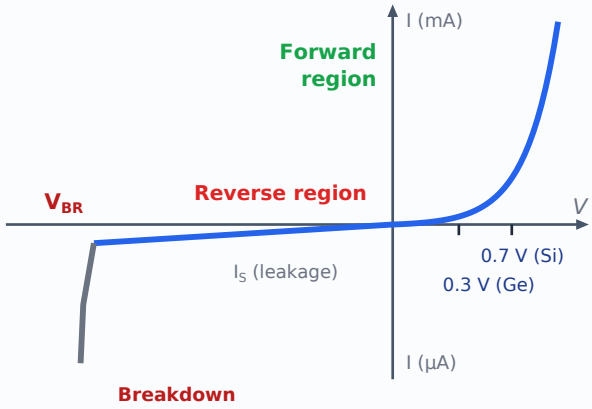


Fig 1.4 V-I characteristic of a PN junction diode (forward and reverse scales differ: mA vs μA)

Diode equation

$$I = I_0 \left(e^{V / \eta V_T} - 1 \right)$$

I_0 = reverse saturation current; $V_T = kT/q \approx 26$ mV at 300 K; η = 1 (Ge), about 2 (Si)

- **Knee (cut-in) voltage:** below it, current is almost zero. Si ≈ 0.7 V, Ge ≈ 0.3 V.
- **Reverse saturation current:** Si in nA, Ge in μA; it rises with temperature (roughly doubles every 10 °C).
- **Breakdown voltage V_{BR}:** a large reverse voltage makes the current rise suddenly and can destroy an ordinary diode.
- **Static resistance** $R_{DC} = V / I$. **Dynamic resistance** $r_d = \Delta V / \Delta I \approx$ **26 mV / I_D** (small in the forward region).

Example: dynamic resistance

A Si diode carries $I_D = 5$ mA. Then $r_d \approx 26$ mV / 5 mA = **5.2 Ω**. At 1 mA it would be 26 Ω, so the diode resists less when it conducts more.

APPLICATIONS

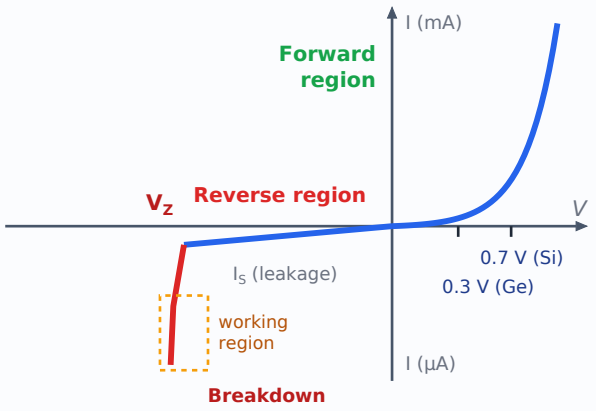
Rectifiers (AC to DC), clippers and clampers, signal detectors, logic gates, reverse-polarity protection, voltage multipliers.

4 Zener Effect and Zener Diode

Zener diode
A specially **heavily doped** PN diode designed to work **safely in the reverse breakdown region**. In breakdown its voltage stays almost **constant (V_Z)** while the current changes widely, so it is a voltage reference/regulator.

Two breakdown mechanisms

	Zener effect	Avalanche effect
Occurs at	Low voltage (V_Z below ~5 V)	Higher voltage (above ~7 V)
Junction	Very heavy doping, very thin depletion layer	Lightly doped, wider depletion layer
Cause	Strong electric field (~10^7 V/m) pulls electrons straight out of covalent bonds (*field emission*)	Fast minority carriers knock out more electrons by *collision*, causing a chain (multiplication)
Temperature coefficient	Negative	Positive



Parameter	Meaning
V_Z	Zener (breakdown) voltage
$I_{Z(min)}$	Minimum current to stay in breakdown
$I_{Z(max)}$	Maximum safe current
$P_{Z(max)}$	$V_Z \times I_{Z(max)}$ (power rating)
Z_Z	Dynamic impedance $\Delta V_Z/\Delta I_Z$ (few Ω)

Fig 1.5 Zener diode characteristic and its main parameters

- **Forward bias:** behaves like an ordinary diode. **Reverse bias:** acts as an open circuit until V_Z, then conducts and holds V_Z.
- **Symbol:** diode symbol with the cathode bar bent like a "Z". Always use a **series resistor** to limit the current: $R_S = (V_{in} - V_Z) / (I_Z + I_L)$.

Example: series resistor for a Zener
Input = 12 V, V_Z = 5.1 V, load current I_L = 20 mA, minimum Zener current I_Z = 5 mA. R_S = (12 − 5.1) / (20 + 5) mA = 6.9 / 25 mA = **276 Ω** (use the nearest standard value, 270 Ω).

APPLICATIONS
Voltage regulators, reference voltage sources, over-voltage protection, waveform clipping, meter protection.

5 Bipolar Junction Transistor (BJT)

DEFINITION
A BJT is a **three-layer, two-junction, three-terminal** device (Emitter, Base, Collector) in which a **small base current controls a large collector current**. "Bipolar" because both electrons and holes take part in conduction.

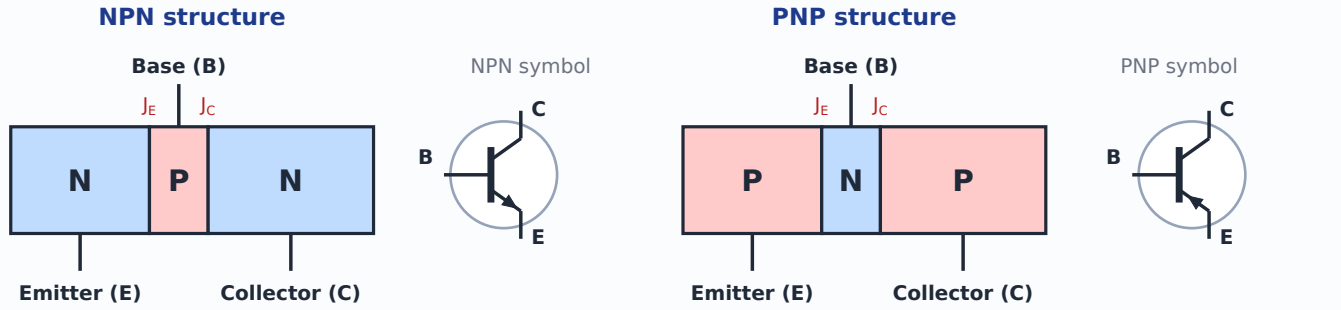


Fig 1.6 Structure and circuit symbols of NPN and PNP transistors (arrow shows direction of conventional emitter current)

Region	Doping	Size	Job
Emitter	Heavy	Medium	Emits (injects) majority carriers
Base	Light	**Very thin**	Controls the carrier flow
Collector	Moderate	**Largest**	Collects the carriers, dissipates heat

Working of an NPN transistor (active mode)

- Emitter-base junction is **forward biased**; collector-base junction is **reverse biased**.
- Electrons from the emitter enter the thin base; only ~1–5 % recombine (this is the small base current I_B).
- The rest are swept into the collector by the reverse field, giving a large $\mathbf{I_C}$.

Mode	EB junction	CB junction	Use
Cut-off	Reverse	Reverse	Switch OFF
Active	**Forward**	**Reverse**	**Amplifier**
Saturation	Forward	Forward	Switch ON

Current relations

$\mathbf{I_E = I_B + I_C}$

$\mathbf{\alpha = I_C / I_E}$ (0.95–0.99) | $\mathbf{\beta = I_C / I_B}$ (20–500)

$\mathbf{\beta = \alpha / (1 - \alpha)}$ | $\mathbf{\alpha = \beta / (1 + \beta)}$ | $\mathbf{I_E = (1 + \beta)\, I_B}$

Example

I_E = 10 mA and I_B = 0.2 mA. Then I_C = 10 − 0.2 = **9.8 mA**, α = 9.8/10 = **0.98**, β = 9.8/0.2 = **49** (check: 0.98/0.02 = 49 ✔).

6 Transistor Configurations: CB, CE and CC

A transistor has three terminals but a two-port network needs four, so **one terminal is made common** to input and output. The common terminal names the configuration.

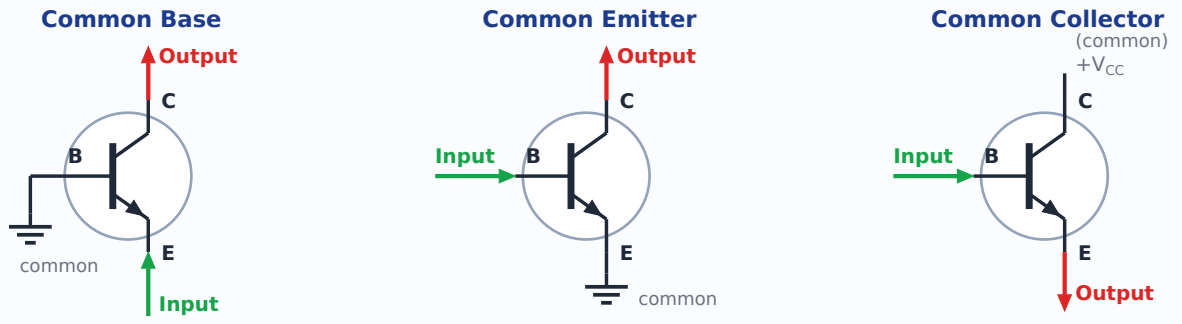


Fig 1.7 Three configurations (green = input, red = output, common terminal is grounded)

Property	CB	CE	CC (emitter follower)
Input terminal / Output	Emitter / Collector	Base / Collector	Base / Emitter
Input resistance	Very low (~50 Ω)	Medium (~1 kΩ)	Very high (~500 kΩ)
Output resistance	Very high (~1 MΩ)	High (~50 kΩ)	Very low (~50 Ω)
Current gain	α (just below 1)	β (high)	1 + β (high)
Voltage gain	High	High	Approx. 1 (slightly below)
Power gain	Medium	**Highest**	Medium
Phase (in to out)	0° (in phase)	**180° (inverted)**	0° (in phase)
Typical use	High-frequency / RF amplifiers	Audio amplifiers, switching	Buffer, impedance matching

EXAM TIP
CE is the most popular configuration because it gives high voltage, current *and* power gain. CC has gain ≈ 1 but is a perfect "buffer" between a high-impedance source and a low-impedance load.

7 Input and Output Characteristics

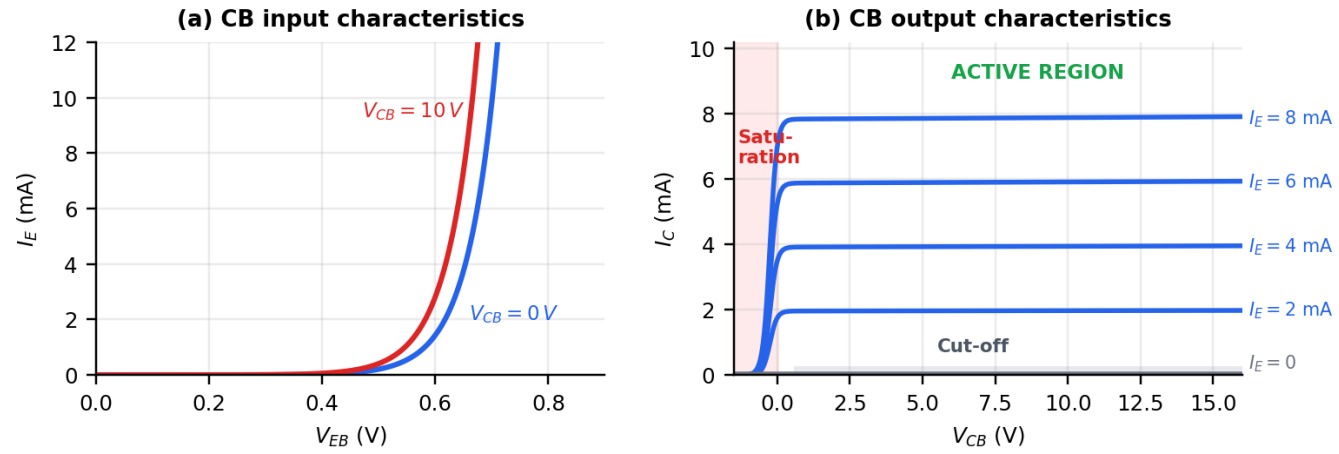


Fig 1.8(a) Common-base input and output characteristics

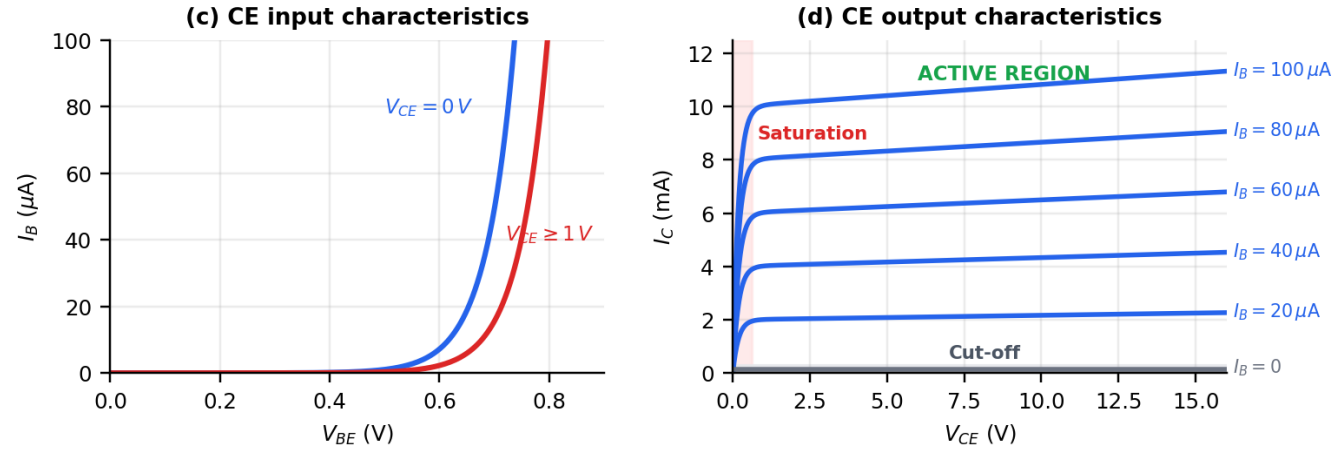


Fig 1.8(b) Common-emitter input and output characteristics

- **CB input (a):** I_E vs V_{EB} at constant V_{CB}. It looks like a forward diode curve; a larger V_{CB} shifts the curve slightly left.
- **CB output (b):** I_C vs V_{CB} at constant I_E. Almost flat lines, because $I_C \approx \alpha I_E$; the curve drops to zero only when V_{CB} is made forward (saturation).
- **CE input (c):** I_B vs V_{BE} at constant V_{CE}; a diode-like curve with a slightly higher cut-in at larger V_{CE}.

- **CE output (d):** I_C vs V_{CE} at constant I_B; $I_C \approx \beta I_B$ and the curves have a gentle upward slope.
- **CC characteristics:** input is I_B vs V_{BC}; output is I_E vs V_{EC}, which is nearly the same as the CE output (since $I_E \approx I_C$).

Region	CE condition	Meaning
Cut-off	$I_B = 0$, $I_C = I_{CEO}$ (tiny)	Transistor OFF; $V_{CE} \approx V_{CC}$
Active	EB forward, CB reverse	Linear amplification; $I_C = \beta I_B$
Saturation	$V_{CE(sat)} \approx 0.2$ V	Transistor fully ON; I_C is set by the external circuit

8 Small-Signal CE Amplifier (Elementary Treatment)

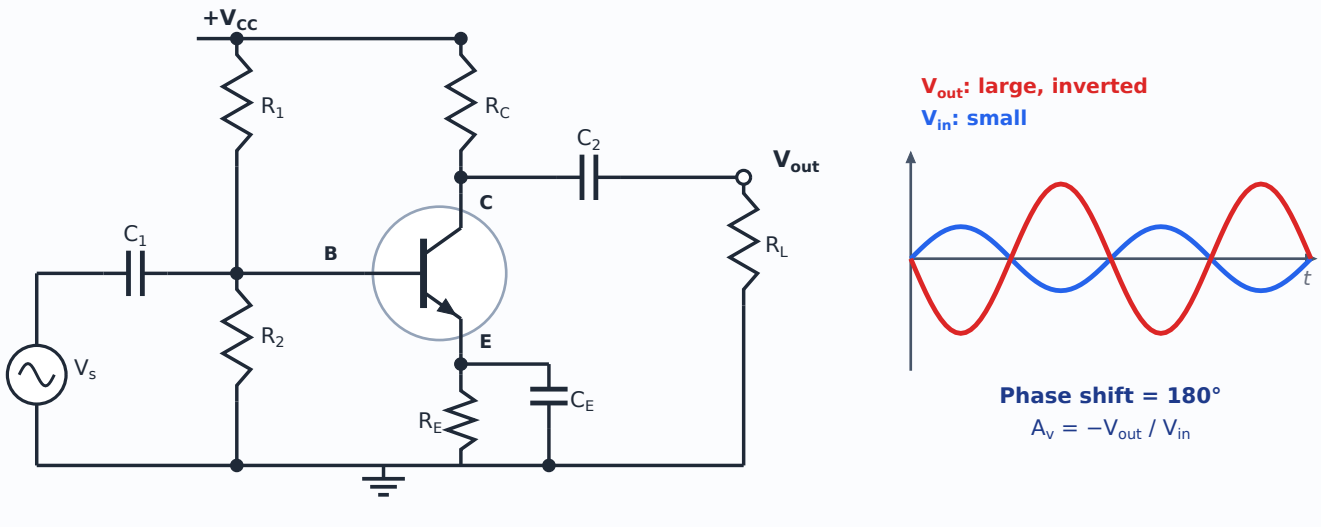


Fig 1.9 Voltage-divider-biased CE amplifier and its input/output waveforms

Role of each component

Component	Purpose
R_1, R_2	Voltage divider that sets a **stable DC bias** (Q-point) at the base
R_C	Converts the collector current change into an output voltage
R_E	Gives **stability** against temperature and β changes (negative feedback)
C_E (bypass)	Short-circuits R_E for AC so the gain is not reduced
C_1, C_2 (coupling)	Pass the AC signal and **block DC** between source, amplifier and load

Working

- **DC:** the biasing network fixes the **Q-point** in the middle of the active region.
- **AC:** in the +ve half-cycle of V_{in}, I_B and I_C increase, so the drop across R_C increases and V_{CE} **decreases**. In the −ve half-cycle the opposite happens.
- So the output is an **amplified and 180° phase-inverted** copy of the input.

Elementary small-signal analysis

DC: $V_B = V_{CC} \cdot R_2/(R_1+R_2)$; $V_E = V_B - 0.7$; $I_C \approx I_E = V_E/R_E$; $V_{CE} = V_{CC} - I_C(R_C+R_E)$

AC: $r_e = 26\text{ mV} / I_E$; $\mathbf{A_v = -(R_C \parallel R_L) / r_e}$; $Z_{in} \approx R_1 \| R_2 \| \beta r_e$; $Z_{out} \approx R_C$

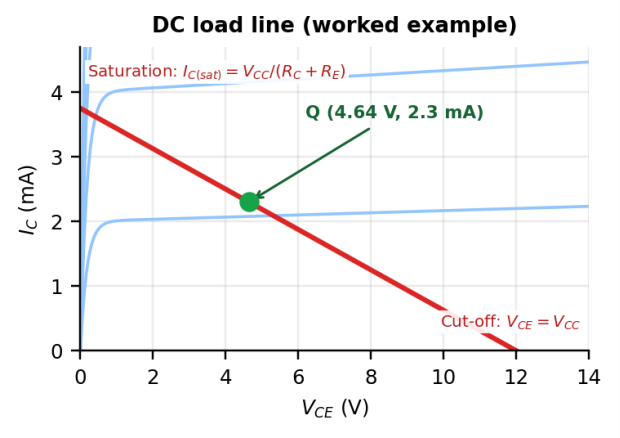
Worked example

$V_{CC} = 12$ V, $R_1 = 30$ kΩ, $R_2 = 10$ kΩ, $R_E = 1$ kΩ, $R_C = 2.2$ kΩ, $R_L = 10$ kΩ.

$V_B = 12 \times 10/40 = 3$ V → $V_E = 2.3$ V → $I_E \approx I_C = 2.3$ mA → $V_{CE} = 12 - 2.3 \times 3.2 =$ **4.64 V** (active ✔).

$r_e = 26/2.3 = 11.3$ Ω; $R_C \| R_L = 1.8$ kΩ; $\mathbf{A_v \approx -1800/11.3 \approx -160}$.

DC load line and Q-point



The **DC load line** joins the two extreme points of the circuit: **cut-off** ($I_C = 0$, $V_{CE} = V_{CC}$) and **saturation** ($V_{CE} \approx 0$, $I_C = V_{CC}/(R_C+R_E)$).

The **Q-point** (quiescent point) is the DC operating point on this line. It must sit near the **middle of the active region** so the signal swings equally up and down without clipping.

Q too high → *saturation clipping*; Q too low → *cut-off clipping*.

Fig 1.10 DC load line and Q-point for the worked example

Transistor as a switch

- **Cut-off** ($I_B = 0$): switch OPEN, $V_{CE} \approx V_{CC}$. • **Saturation** (large I_B): switch CLOSED, $V_{CE(sat)} \approx 0.2$ V.
- Used as the basic element of logic gates, relay/LED drivers and memory cells (links to Unit III).

9 Quick Summary and Exam Corner

Unit I in one glance

- Vacuum tube → transistor (1947) → IC (1958) → VLSI → nano-electronics: smaller, faster, cheaper, cooler.
- PN diode: forward = ON (knee 0.7 V Si, 0.3 V Ge); reverse = OFF with tiny leakage; breakdown at V_{BR}.
- Zener diode: operated in reverse breakdown, holds a constant V_Z; Zener effect (low V) and avalanche (high V).
- BJT: $I_E = I_B + I_C$; $\beta = \alpha/(1-\alpha)$. Modes: cut-off, active (amplifier), saturation.
- CE: highest power gain and 180° phase shift; CB: low R_{in}, good at high frequency; CC: gain ≈ 1, buffer.
- CE amplifier: R_1R_2 bias, R_E stability, C_E bypass, C_1C_2 coupling; $A_v = -R_C \| R_L / r_e$.

Short answer questions (2 marks)

1. What is the barrier potential of Si and Ge diodes?
2. Define dynamic resistance of a diode.
3. Distinguish Zener and avalanche breakdown (any two points).
4. Why is the base of a transistor made thin and lightly doped?
5. Define α and β and write the relation between them.
6. Why is CE the most widely used configuration?

Long answer questions (5 / 10 marks)

1. Explain the forward and reverse V-I characteristics of a PN junction diode with a neat diagram.
2. Explain the Zener diode characteristics and the Zener and avalanche breakdown mechanisms.
3. Draw the CE configuration and explain its input and output characteristics, marking the three regions.
4. Compare CB, CE and CC configurations in a table.
5. With a circuit diagram, explain the working of a small-signal CE amplifier. Derive A_v and state the role of each component.
6. Trace the evolution of electronics from vacuum tubes to nano-electronics.

Numerical practice (with answers)

1. A transistor has $I_B = 40$ µA and $\beta = 100$. Find I_C and I_E. *Ans: $I_C = 4$ mA, $I_E = 4.04$ mA*
2. Find β if $\alpha = 0.99$. *Ans: $\beta = 0.99/0.01 = 99$*
3. A Zener diode ($V_Z = 6.2$ V, $P_{Z(max)} = 0.5$ W). Find $I_{Z(max)}$. *Ans: 0.5/6.2 ≈ 80.6 mA*
4. For a CE amplifier, $I_E = 1.3$ mA, $R_C \| R_L = 2$ kΩ. Find A_v. *Ans: $r_e = 20\ \Omega$, $A_v = -100$*

Common mistakes to avoid

- Mixing up the biasing: **forward** = P to positive; **reverse** = P to negative.
- Saying “transistor amplifies current only”: CE amplifies **voltage, current and power**.
- Forgetting the **series resistor** with a Zener diode, or forgetting that CE **inverts** the signal.
- Drawing characteristic curves without labelling axes, constant parameter and the three regions.

Quick check: pick the right answer

1. The depletion region **widens** in: (a) forward bias (b) reverse bias (c) both (d) none. *Ans: (b)*
2. The phase shift of a CE amplifier is: (a) 0° (b) 90° (c) 180° (d) 360°. *Ans: (c)*
3. Emitter follower is another name for: (a) CB (b) CE (c) CC. *Ans: (c)*
4. A transistor used as an amplifier operates in the: (a) cut-off (b) active (c) saturation region. *Ans: (b)*
5. Zener breakdown below about 5 V is mainly due to: (a) collision (b) field emission (c) diffusion. *Ans: (b)*
6. Which configuration has the highest input resistance? (a) CB (b) CE (c) CC. *Ans: (c)*

BASIC ELECTRONICS ENGINEERING • PART B

UNIT II

Basic Electronic Circuits & Instrumentation

Rectifiers, filters, regulators, public address system, RC-coupled amplifier and instrumentation

WHAT YOU WILL LEARN

1 Block diagram of a DC power supply
2 Half-wave & full-wave bridge rectifier
3 Capacitor filter
4 Zener voltage regulator
5 Public Address (PA) system
6 RC-coupled CE amplifier & response
7 Electronic instrumentation system
8 Summary & exam questions

1 DC Power Supply: Block Diagram

Most electronic circuits need **steady DC**, but the wall socket gives **AC**. A DC power supply converts one into the other in five simple stages.

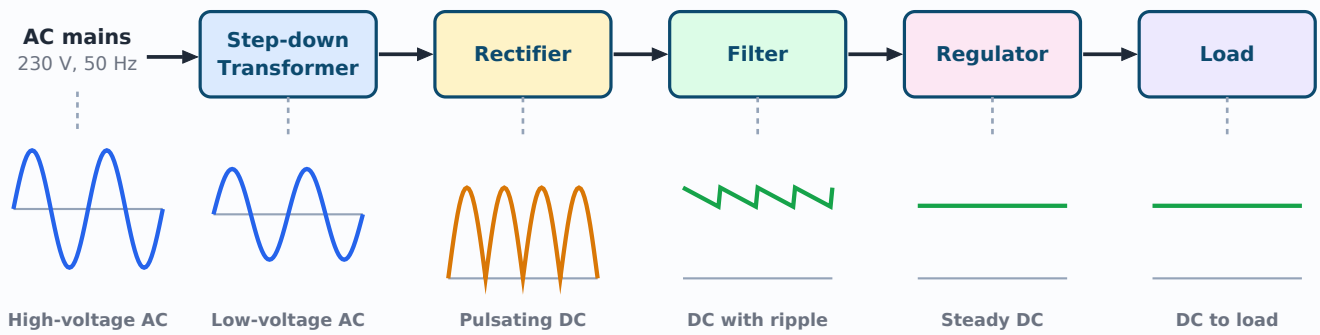


Fig 2.1 Block diagram of a regulated DC power supply, with the waveform at each stage

Block	Function
Step-down transformer	Reduces the 230 V AC mains to a safe, low voltage (e.g. 12 V) and gives electrical isolation
Rectifier	Converts AC to **pulsating DC** using diodes (they conduct in one direction only)
Filter	Smooths the pulsations (removes most of the ripple), usually using a capacitor
Regulator	Keeps the output voltage **constant** despite changes in mains voltage or load (e.g. Zener / IC 7805)
Load	The electronic circuit or device being powered

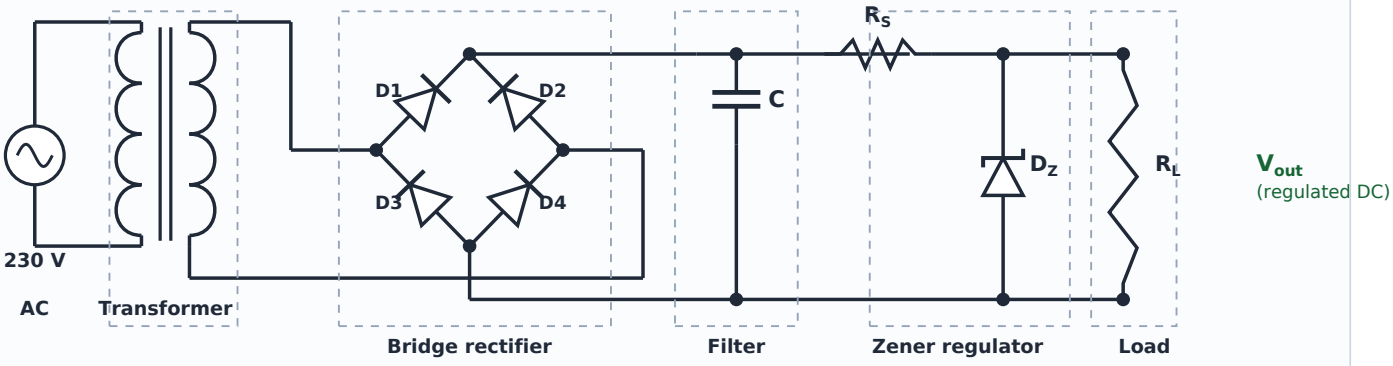


Fig 2.2 Complete regulated power supply circuit (transformer, bridge rectifier, capacitor filter and Zener regulator)

2 Rectifiers: Half-Wave and Full-Wave Bridge

Starting point: the half-wave rectifier

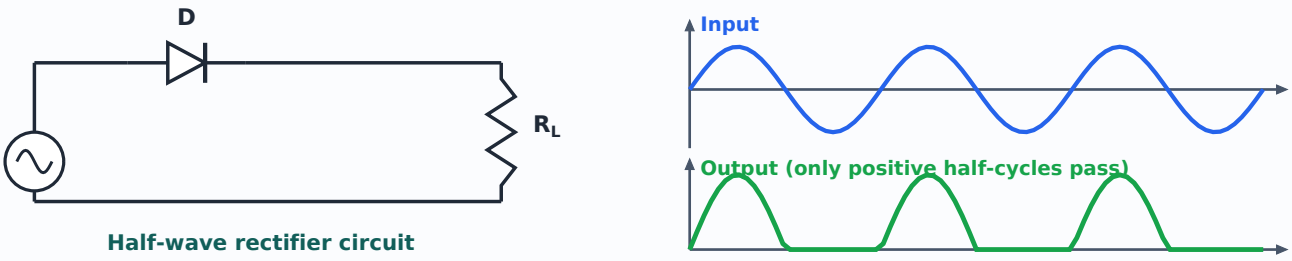


Fig 2.3 Half-wave rectifier circuit and waveforms

- One diode conducts only in the **positive half-cycle**; during the negative half-cycle it is reverse biased and the output is zero.
- $\mathbf{V_{dc} = V_m/\pi \approx 0.318\ V_m}$; ripple factor 1.21; efficiency 40.6 %; PIV = V_m; ripple frequency = f.
- Drawback: half of the input is wasted and the output is very rough, so we use **full-wave** rectification.

Full-wave bridge rectifier

DEFINITION
A **rectifier** converts AC into unidirectional (pulsating DC) current. The **bridge rectifier** uses **four diodes** in a diamond so that **both half-cycles** of the AC input reach the load in the **same direction**, and it needs no centre-tapped transformer.

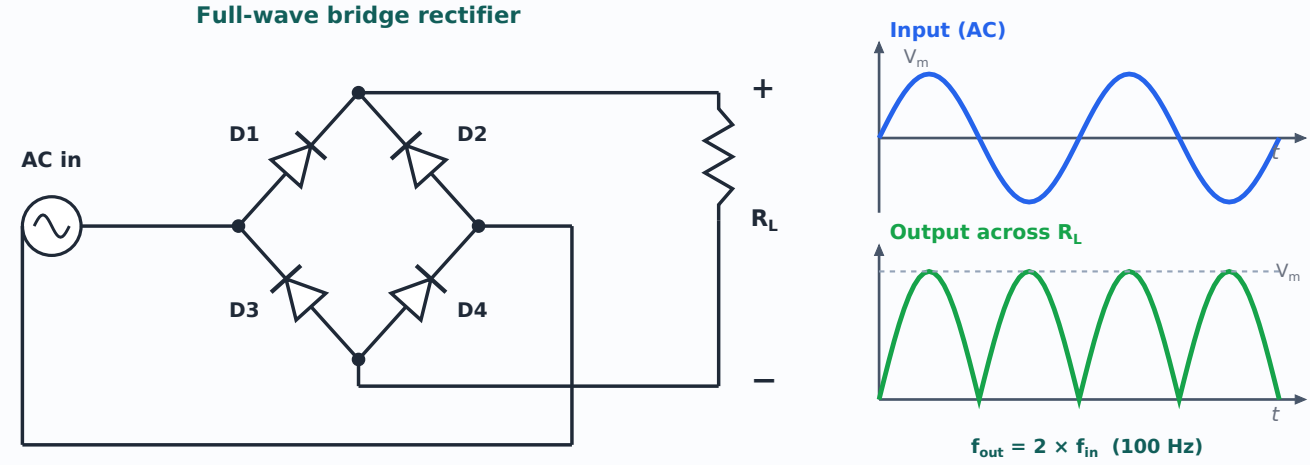


Fig 2.4 Bridge rectifier circuit and its input/output waveforms

Working

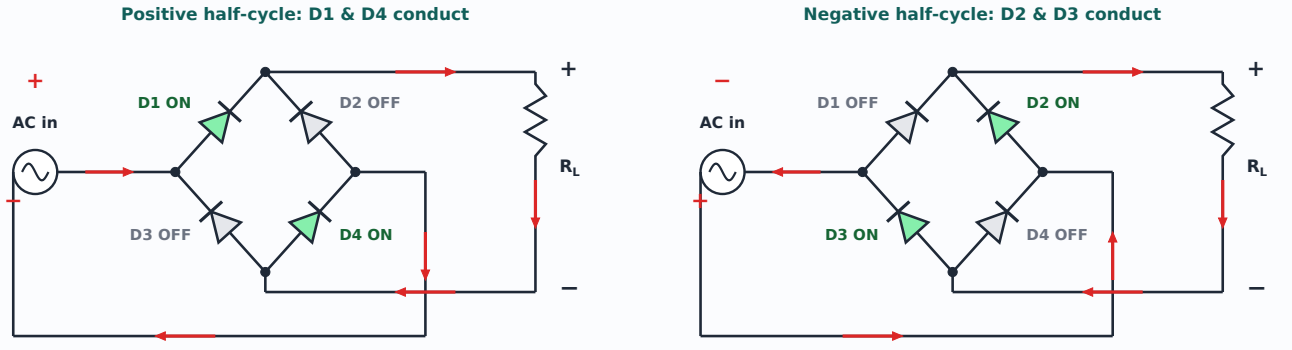


Fig 2.5 Current path in the two half-cycles (red arrows: direction of current; green diodes: conducting)

Input half-cycle	Diodes ON	Diodes OFF	Current through R_L
Positive (top end +)	D_1 and D_4	D_2 and D_3	Top to bottom
Negative (top end −)	D_2 and D_3	D_1 and D_4	**Top to bottom (same direction)**

Important formulas (ideal diodes, peak secondary voltage V_m)
$V_{dc} = 2V_m/\pi \approx 0.637\ V_m$ | $V_{rms} = V_m/\sqrt{2}$ | $I_{dc} = 2I_m/\pi$
Ripple factor $\gamma \approx 0.48$ | Efficiency $\eta = 81.2\ \%$ | PIV $= V_m$ | Ripple frequency $= 2f$

Parameter	Half-wave	Centre-tap full-wave	Bridge full-wave
Number of diodes	1	2	4
V_{dc}	V_m/π	$2V_m/\pi$	$2V_m/\pi$
PIV of each diode	V_m	$2V_m$	**V_m**
Ripple factor	1.21	0.48	0.48
Efficiency	40.6 %	81.2 %	81.2 %
Ripple frequency	f	2f	2f
Centre-tap transformer	No	**Yes**	**No**

Performance term	Meaning
Ripple factor (γ)	Ratio of AC (ripple) component to DC component in the output; smaller is better ($\gamma = V_{r(rms)} / V_{dc}$)
PIV (peak inverse voltage)	Maximum reverse voltage that a diode must withstand without breaking down
Efficiency (η)	DC output power / AC input power × 100 %
Regulation	How little the output voltage changes with load current (and input voltage)

Example
Secondary = 12 V rms. $V_m = 12 \times 1.414 = 16.97$ V. Ideal $V_{dc} = 0.637 \times 16.97 \approx$ **10.8 V**.
In practice two diodes (2 × 0.7 V = 1.4 V) conduct at a time: $V_{dc} \approx 0.637 \times (16.97 - 1.4) \approx$ **9.9 V**. Each diode needs PIV ≥ 17 V (a 1N4007 is ample).

Why the bridge is preferred
- Uses the whole transformer winding all the time (better utilisation), PIV across each diode is only V_m.
- Output ripple is at 2f (100 Hz), which is easier to filter than f (50 Hz) in a half-wave rectifier.
- Disadvantage: two diode drops (≈1.4 V) in the path, so a little more power loss at low voltages.

3 Capacitor Filter

The rectifier output is DC in direction but not steady: it pulses at 100 Hz. A **capacitor filter** (a large electrolytic capacitor connected **in parallel** with the load) stores charge at the peaks and releases it in the gaps.

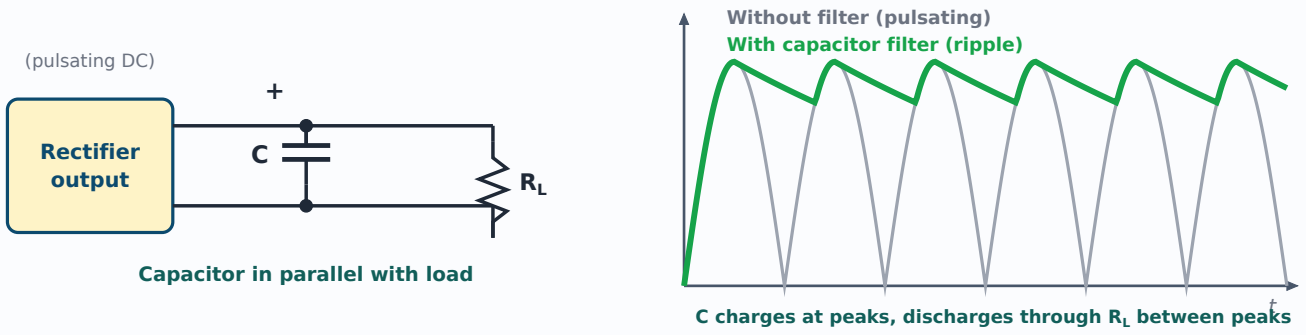


Fig 2.6 Capacitor filter and the smoothed output voltage

- **Rising part:** when the rectified voltage rises above the capacitor voltage, the diodes conduct and the capacitor **charges** quickly to the peak V_m.
- **Falling part:** the input drops below the capacitor voltage, the diodes switch OFF and the capacitor **discharges slowly** through R_L (time constant $R_L C$) so the load voltage barely falls.
- **Next peak:** the capacitor is topped up again, which gives a nearly flat DC with a small **ripple**.

Effects of the capacitor filter

- Larger C or larger R_L (lighter load) → **less ripple** and higher V_{dc} (closer to V_m).
- Heavy load (small R_L) → capacitor drains faster → **more ripple**.
- Typical values: 470 µF to 4700 µF electrolytic; observe correct polarity and a voltage rating above V_m.

4 Simple Zener Voltage Regulator

Even after filtering, the DC output changes when the mains voltage or the load changes. A **regulator** keeps V_{out} constant. The simplest one uses a **Zener diode in reverse breakdown** in parallel with the load, plus a series resistor R_S.

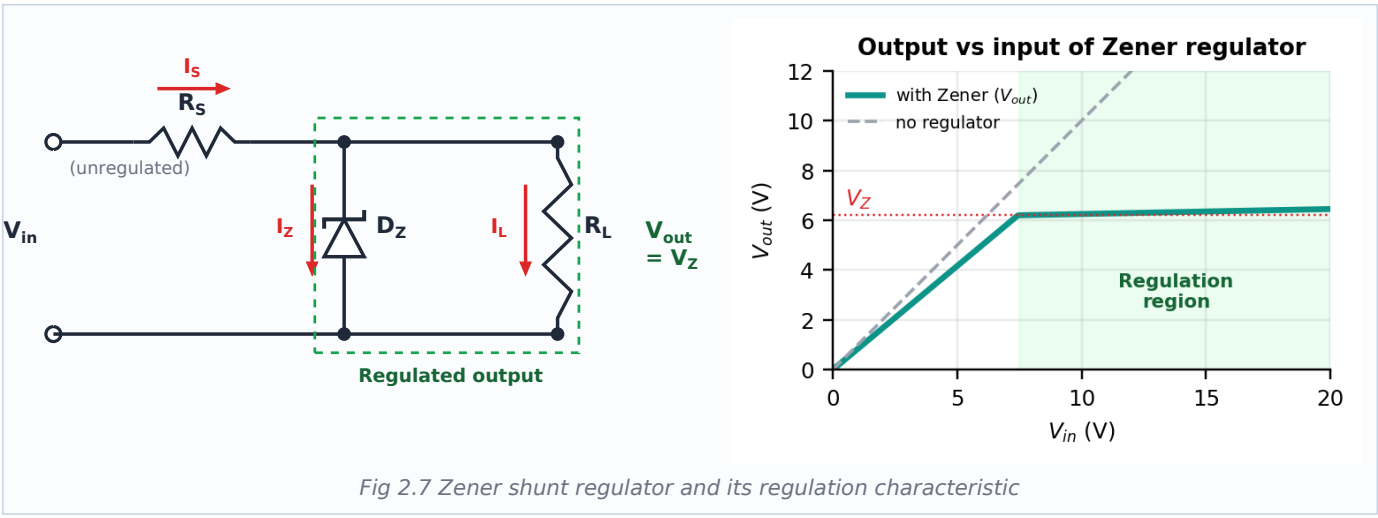


Fig 2.7 Zener shunt regulator and its regulation characteristic

Working

- The Zener diode is **reverse biased** and operates in breakdown, so the voltage across it (and the load) stays at $\mathbf{V_Z}$.
- **If V_{in} increases:** I_S rises; the extra current flows through the Zener (I_Z ↑). The extra drop appears across R_S, so V_{out} stays at V_Z.
- **If load current I_L increases:** I_Z decreases by the same amount, I_S stays almost constant, so V_{out} stays at V_Z.
- **Rule:** I_Z must stay between $I_{Z(min)}$ and $I_{Z(max)}$ for regulation to hold.

Design equations

$\mathbf{I_S = I_Z + I_L = (V_{in} - V_Z) / R_S}$ | $\mathbf{V_{out} = V_Z}$ | $\mathbf{P_{Z(max)} = V_Z\, I_{Z(max)}}$

Line regulation = $\Delta V_{out}/\Delta V_{in} \times 100\ \%$ | Load regulation = $(V_{NL} - V_{FL})/V_{FL} \times 100\ \%$

Example

V_{in} = 12 V, V_Z = 5.1 V, R_L = 1 kΩ → I_L = 5.1 mA. For I_Z = 5 mA: $R_S = (12 - 5.1)/(5 + 5.1)$ mA = **683 Ω ≈ 680 Ω**.

Limitations

Wastes power in R_S and the Zener, fixed output, suitable only for small load currents. Higher currents use transistor or IC regulators (e.g. 7805).

5 Amplifiers and the Public Address (PA) System

DEFINITION

An **amplifier** increases the strength (voltage, current or power) of a signal without changing its shape. Voltage gain $A_v = V_o/V_i$; in decibels, **gain (dB) = 20 $\log_{10} A_v$**.

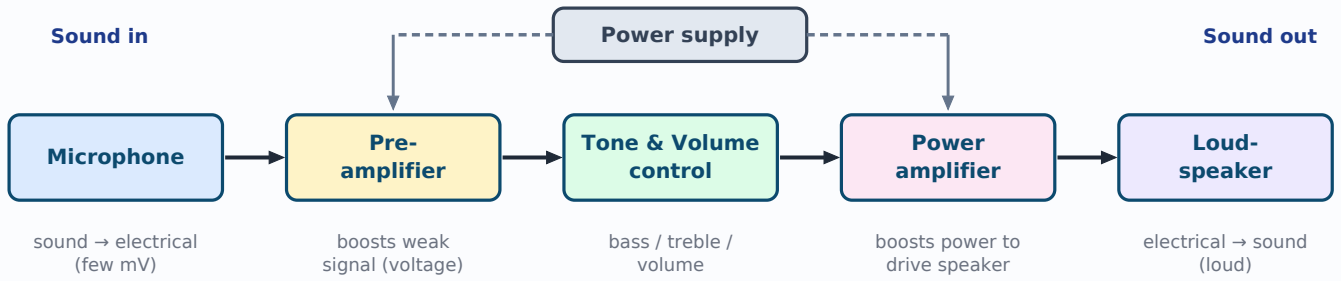


Fig 2.8 Block diagram of a Public Address (PA) system

Block	Role
Microphone	Transducer: converts sound waves into a tiny electrical signal (a few mV)
Pre-amplifier	Low-noise voltage amplifier that raises the weak signal to a usable level
Tone / volume control	Adjusts bass, treble and loudness; shapes the signal before power amplification
Power amplifier	Supplies high current/power needed to drive the loudspeaker
Loudspeaker	Transducer: converts the electrical signal back to sound
Power supply	Gives DC to the amplifier stages

Voltage gain A_v	1	2	10	100	1000	0.707
Gain in dB	0	6 dB	20 dB	40 dB	60 dB	−3 dB

Acoustic feedback (“howling”)

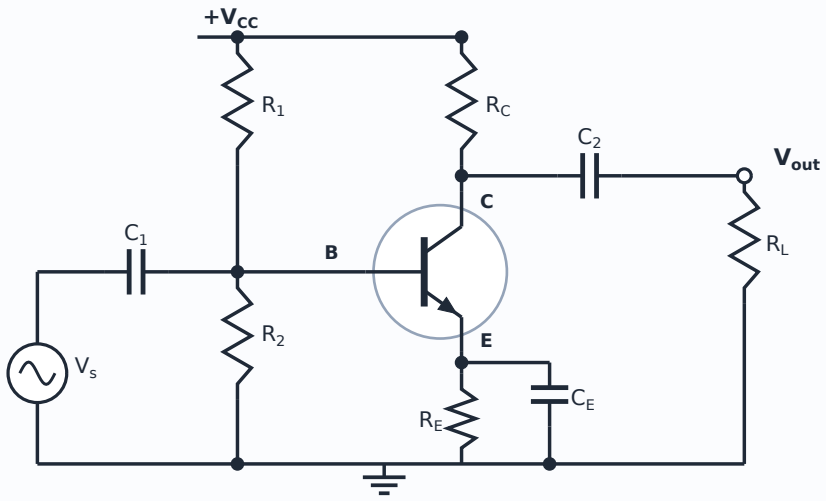
If the loudspeaker sound re-enters the microphone, it is amplified again and again, producing a loud whistle. Cure: keep the microphone away from the speakers, lower the gain or use directional microphones.

APPLICATIONS

Railway stations, airports, auditoriums, classrooms, stadiums, religious places, emergency announcements.

6 RC-Coupled CE Amplifier and its Frequency Response

When one transistor stage cannot give enough gain, stages are cascaded. They are linked by a **coupling network**; the most common is **RC coupling**, where a capacitor passes the AC signal and blocks the DC of the previous stage.



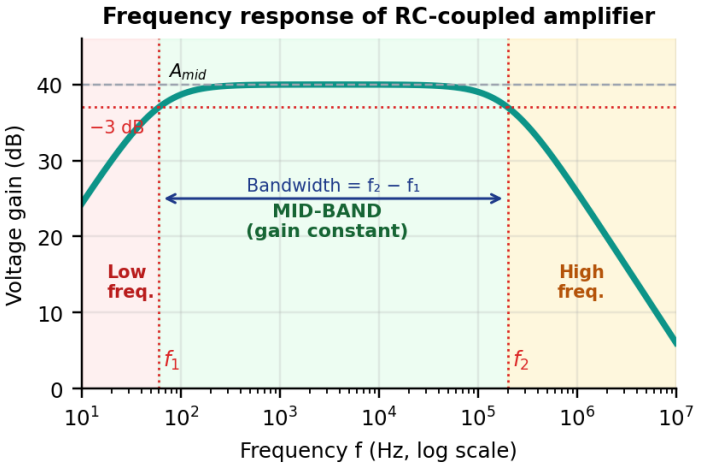
Components

- **R_1, R_2** – voltage-divider bias
- **R_C** – collector (load) resistor
- **R_E** – bias stabilisation
- **C_E** – bypass: prevents gain loss
- **C_1, C_2** – coupling capacitors: pass AC, block DC
- **R_L** – next stage / load

Fig 2.9 Single stage of an RC-coupled common-emitter amplifier

Working

- **DC:** R_1, R_2, R_E fix the Q-point in the active region, so the transistor always conducts.
- **AC:** the input signal passes through C_1 to the base and varies I_B. The collector current varies β times more, producing a large signal voltage across R_C.
- The amplified signal, **180° out of phase** with the input, goes through C_2 to the load. C_2 blocks the DC at the collector.



Why does the gain change with frequency?

Low frequencies: reactance $X_C = 1/(2\pi fC)$ of C_1, C_2 and C_E is large, so part of the signal is dropped across them → gain falls.

Mid frequencies: capacitors behave like short circuits; transistor capacitances are negligible → **gain is maximum and constant**.

High frequencies: transistor junction and stray capacitances shunt the signal, and β falls → gain falls.

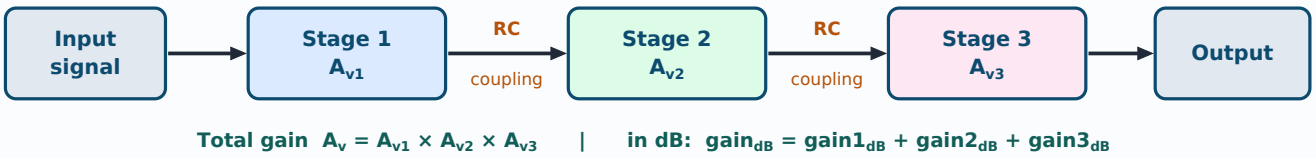
Fig 2.10 Frequency response curve (gain in dB against log frequency)

Bandwidth

The **half-power (−3 dB) frequencies** f_1 (lower) and f_2 (upper) are where the gain drops to **0.707 A_{mid}** ($A_{mid}/\sqrt{2}$). **Bandwidth BW = $f_2 - f_1$**. For audio amplifiers this is about 20 Hz to 20 kHz.

Advantages	Disadvantages
Simple, cheap and compact; no bulky transformers or inductors	Poor impedance matching, so low power transfer to a low-resistance load (e.g. a speaker)
Good, wide, flat response over the audio range	Gain falls at very low and very high frequencies
Little frequency distortion; easy to cascade many stages	Cannot amplify DC; power gain is modest (so used for voltage amplification)

Multistage amplifiers and coupling methods



Total gain $A_v = A_{v1} \times A_{v2} \times A_{v3}$ | **in dB:** $gain_{dB} = gain1_{dB} + gain2_{dB} + gain3_{dB}$

Fig 2.11 Cascaded amplifier stages joined by coupling networks

Coupling	How it works	Remark
RC coupling	Capacitor + resistor pass AC and block DC	Cheap, wide bandwidth: audio voltage amplifiers
Transformer coupling	Transformer transfers the signal and matches impedance	Good power transfer but bulky, costly; narrow band
Direct coupling	Output of one stage connected directly to the next	Can amplify DC; stage biasing interacts (drift)

7 Electronic Instrumentation System

DEFINITION

Electronic instrumentation is the use of electronic circuits to **measure, display, record and control** physical quantities (temperature, pressure, light, force, speed, heart-beat, ...) by first converting them into electrical signals.

Example: digital thermometer → heat ▶ thermistor ▶ amplifier ▶ ADC ▶ numeric display

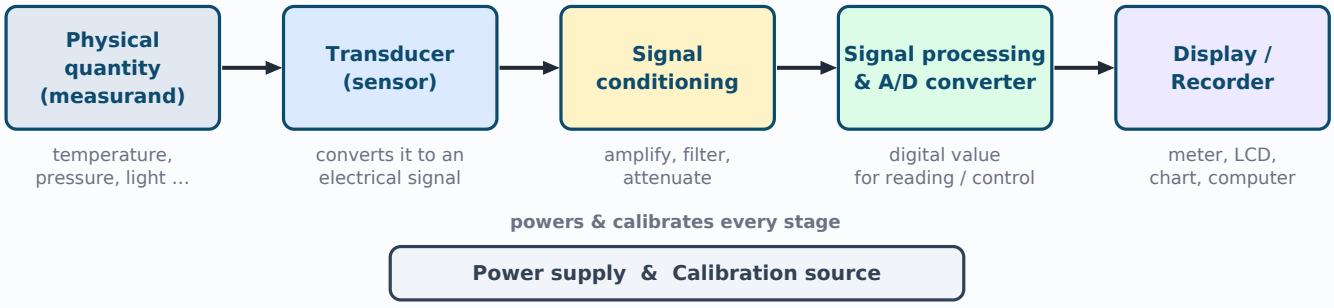


Fig 2.12 Block diagram of an electronic instrumentation system

Block	Function	Example
Transducer / sensor	Converts a physical quantity into an electrical signal	Thermocouple, thermistor, LDR, strain gauge, microphone
Signal conditioning	Amplifies, filters, attenuates or linearises the weak signal	Op-amp amplifier, low-pass filter
Signal processing / ADC	Converts analog to digital; computes, compares, controls	Microcontroller, ADC chip
Display / recorder	Presents the result for the user or stores/transmits it	LCD, LED, meter, chart recorder, PC
Power supply, calibration	Powers all blocks; reference to keep accuracy	Regulated DC supply

Common transducers

Quantity	Transducer	Principle
Temperature	Thermistor, thermocouple, RTD	Resistance / voltage changes with temperature
Light	LDR, photodiode	Resistance / current changes with light intensity
Pressure, force, weight	Strain gauge, load cell	Resistance changes when stretched
Displacement	LVDT, potentiometer	Induced voltage / resistance varies with position
Sound	Microphone	Sound pressure to voltage

Active vs passive transducers

- **Active** transducers generate their own voltage (thermocouple, piezo-electric crystal, photovoltaic cell).
- **Passive** transducers need an external supply (thermistor, strain gauge, LDR).

APPLICATIONS

Digital multimeter, oscilloscope, ECG / patient monitor, weighing machine, industrial process control, weather stations, smart-home sensors.

Advantages of electronic instruments

High sensitivity and speed, remote measurement, easy digital display and storage, small size, and the ability to connect to computers and control systems.

8 Quick Summary and Exam Corner

Unit II in one glance

- DC supply = **Transformer → Rectifier → Filter → Regulator → Load**.
- Bridge rectifier: 4 diodes; two conduct in each half-cycle; $V_{dc} = 2V_m/\pi$; PIV = V_m; ripple frequency = 2f; $\eta = 81.2$ %.
- Capacitor filter: C in parallel with load; charges at peaks, discharges slowly; larger C → smaller ripple.
- Zener regulator: reverse-biased Zener in parallel with load plus R_S; $V_{out} = V_Z$; $R_S = (V_{in} - V_Z)/(I_Z + I_L)$.
- PA system: Microphone → Pre-amp → Tone control → Power amp → Loudspeaker.
- RC-coupled CE amplifier: flat mid-band; gain falls at low f (coupling/bypass caps) and high f (junction/stray caps); BW = $f_2 - f_1$.
- Instrumentation: Measurand → Transducer → Signal conditioning → Processing/ADC → Display.

Short answer questions (2 marks)

1. List the blocks of a regulated DC power supply.
2. Why is a bridge rectifier preferred over a centre-tap rectifier?
3. What is the ripple frequency of a full-wave rectifier with 50 Hz input?
4. State the function of a filter and of a regulator.
5. What is the purpose of the bypass capacitor in a CE amplifier?
6. Define bandwidth of an amplifier.
7. What is a transducer? Give two examples.

Long answer questions (5 / 10 marks)

1. Draw the block diagram of a DC power supply and explain each block with waveforms.
2. With a neat circuit diagram, explain the working of a full-wave bridge rectifier. Give its V_{dc}, ripple factor, efficiency and PIV.
3. Explain the working of a capacitor filter with waveforms.
4. Explain the working of a Zener voltage regulator for variations in input voltage and load.
5. Draw the block diagram of a PA system and explain the function of each block.
6. Draw the circuit of an RC-coupled CE amplifier, explain its working and sketch its frequency response.

7. Draw and explain the block diagram of an electronic instrumentation system.

Numerical practice (with answers)

1. A bridge rectifier has a secondary of 20 V rms. Find V_m, ideal V_{dc} and PIV. *Ans: V_m = 28.3 V, V_{dc} = 18.0 V, PIV = 28.3 V*
2. A Zener (V_Z = 9.1 V) is fed from 15 V through R_S = 590 Ω. With no load, find the Zener current. *Ans: (15 − 9.1)/590 = 10 mA*
3. An amplifier has A_{mid} = 200, f_1 = 40 Hz, f_2 = 80 kHz. Find the bandwidth and the gain at the −3 dB points. *Ans: BW ≈ 79.96 kHz; gain = 200/√2 ≈ 141*
4. Convert a voltage gain of 100 into dB. *Ans: 20 log 100 = 40 dB*

BASIC ELECTRONICS ENGINEERING • PART B

UNIT III

Digital Electronics

Number systems, codes, logic gates, Boolean algebra, adders, flip-flops, registers and counters

WHAT YOU WILL LEARN

1 Number systems & conversions
2 BCD, Excess-3, Gray & Hamming codes
3 Logic gates & universal gates
4 Boolean algebra & theorems
5 Half adder & full adder
6 Sequential circuits & flip-flops
7 Registers and counters
8 Summary & exam questions

1 Overview of Number Systems

Digital circuits work with only two voltage levels, **HIGH (1)** and **LOW (0)**, so they naturally use the **binary** number system. Octal and hexadecimal are shorthand ways to write long binary numbers.

System	Base	Digits used	Example
Decimal	10	0–9	$(45)_{10}$
Binary	2	0, 1	$(101101)_{2}$
Octal	8	0–7	$(55)_{8}$
Hexadecimal	16	0–9, A–F (A=10 … F=15)	$(2D)_{16}$

Dec	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Bin	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
Oct	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17
Hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

Conversion methods

Conversion	Method	Example
Decimal → Binary	Divide repeatedly by 2; read remainders **bottom to top**	45 → 22 r1, 11 r0, 5 r1, 2 r1, 1 r0, 0 r1 → **101101**
Binary → Decimal	Add the weights (…8, 4, 2, 1 . ½, ¼, ⅛)	1101.101 = 8+4+1+0.5+0.125 = **13.625**
Binary → Octal	Group bits in **3s** from the binary point	101 110 011 → $\mathbf{563}_{8}$
Binary → Hex	Group bits in **4s**	1011 1100 → $\mathbf{BC}_{16}$
Hex → Decimal	Multiply digits by powers of 16	2F = 2×16 + 15 = **47**
Decimal → Hex	Divide repeatedly by 16	255 → F, F → **FF**

Binary arithmetic

Addition: 0+0 = 0, 0+1 = 1, 1+0 = 1, **1+1 = 10** (sum 0, carry 1). **Subtraction** is done by adding the **2's complement**.

1's complement = flip every bit; **2's complement** = 1's complement + 1.

Example (4-bit) 7 − 5: 0111 + 2's comp(0101) = 0111 + 1011 = 1 0010 → discard the carry → **0010 = 2** ✓

2 Digital Codes: BCD, Excess-3, Gray and Hamming

A **code** represents numbers or characters as groups of bits. Different codes are chosen for ease of calculation, error reduction or error correction.

Decimal	Binary	BCD (8421)	Excess-3	Gray
0	0000	0000	0011	0000
1	0001	0001	0100	0001
2	0010	0010	0101	0011
3	0011	0011	0110	0010
4	0100	0100	0111	0110
5	0101	0101	1000	0111
6	0110	0110	1001	0101
7	0111	0111	1010	0100
8	1000	1000	1011	1100
9	1001	1001	1100	1101

BCD (Binary Coded Decimal)

Each decimal **digit** is written separately as a 4-bit binary number (weights 8-4-2-1). Codes 1010–1111 are **invalid**.

Example: $(47)_{10}$ = **0100 0111**$_{BCD}$. Used in digital clocks, calculators and meters.

Excess-3 code

BCD + 0011 (3) for every digit. It is **self-complementing** (9's complement = flip all bits), useful in decimal arithmetic.

Example: 5 → 0101 + 0011 = **1000**.

Gray code (reflected binary)

Only one bit changes between consecutive numbers, which avoids glitches in encoders and position sensors.

Binary → Gray: keep the MSB; each next Gray bit = XOR of that binary bit and the one before it. 1011 → 1, 1⊕0=1, 0⊕1=1, 1⊕1=0 → **1110**.

Gray → Binary: keep the MSB; each next binary bit = previous binary bit ⊕ current Gray bit. 1110 → 1, 1⊕1=0, 0⊕1=1, 1⊕0=1 → **1011**.

Hamming code (error detection and correction)

- A **Hamming (7,4)** code adds **3 parity bits** to 4 data bits so that a **single-bit error** can be found *and corrected*.
- Bit positions 1 to 7: parity bits sit at positions **1, 2, 4** (powers of 2); data bits at 3, 5, 6, 7.
- Each parity bit checks a group (even parity): **P1** → positions 1,3,5,7; **P2** → 2,3,6,7; **P4** → 4,5,6,7.

Step	Working for data 1011 (even parity)
Place data	Positions 3,5,6,7 = **1, 0, 1, 1**
P1 (1,3,5,7)	bits 3,5,7 = 1,0,1 → two 1s → **P1 = 0**
P2 (2,3,6,7)	bits 3,6,7 = 1,1,1 → three 1s → **P2 = 1**
P4 (4,5,6,7)	bits 5,6,7 = 0,1,1 → two 1s → **P4 = 0**
Code word	positions 1…7 → **0 1 1 0 0 1 1**
Error check	If received 0110**1**11 (bit 5 flipped): C1 = 1, C2 = 0, C4 = 1 → C4C2C1 = 101 = 5 → **bit 5 is wrong; flip it**

3 Logic Gates and Universal Gates

A **logic gate** is the basic building block of digital circuits: it takes one or more binary inputs and produces one binary output according to a logical rule.

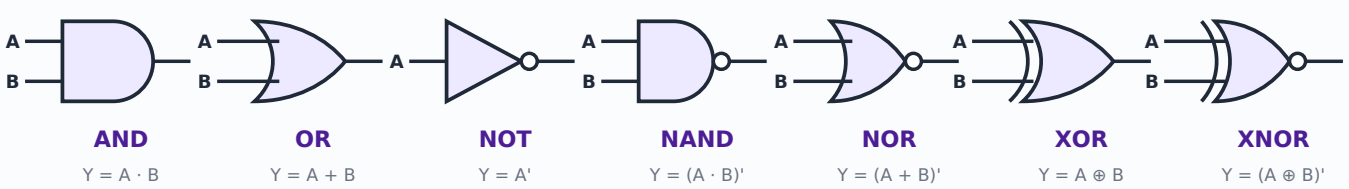


Fig 3.1 Symbols and Boolean expressions of the basic logic gates

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

Gate	Functionality (in words)
NOT (inverter, 1 input: 0→1, 1→0)	Output is the **opposite** of the input
AND	Output is 1 **only when all** inputs are 1
OR	Output is 1 when **at least one** input is 1
NAND / **NOR**	AND / OR followed by NOT (output is the complement)
XOR	Output is 1 when the inputs are **different** (odd number of 1s)
XNOR	Output is 1 when the inputs are the **same** (equality detector)

Universal gates
NAND and **NOR** are called **universal gates** because **any** logic function (NOT, AND, OR and hence all circuits) can be built using only NAND gates or only NOR gates. They are cheap, fast and easy to fabricate.

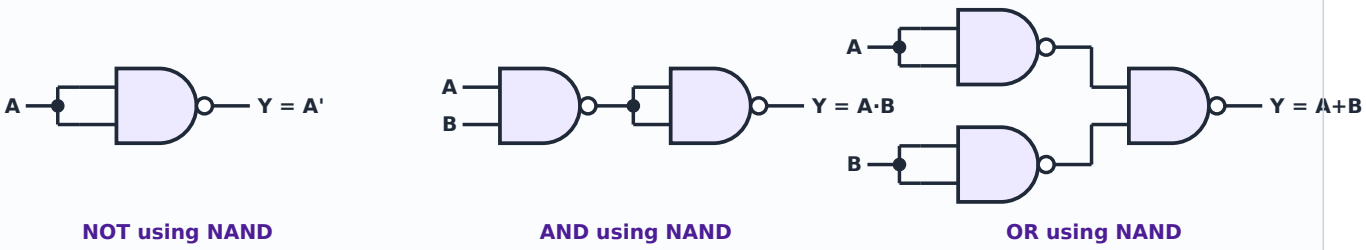


Fig 3.2 NOT, AND and OR gates built using only NAND gates

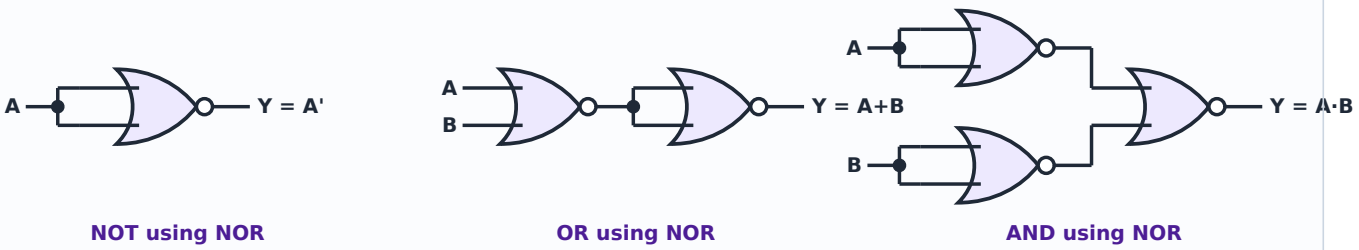


Fig 3.3 NOT, OR and AND gates built using only NOR gates

4 Boolean Algebra: Properties and Theorems

Boolean algebra is the algebra of two-valued variables (0 and 1) with the operations AND (·), OR (+) and NOT (A′ means "NOT A"). It is used to **simplify** logic circuits.

Law	AND form	OR form
Identity	A · 1 = A	A + 0 = A
Null (domination)	A · 0 = 0	A + 1 = 1
Idempotent	A · A = A	A + A = A
Complement	A · A′ = 0	A + A′ = 1
Involution	(A′)′ = A	(A′)′ = A
Commutative	A · B = B · A	A + B = B + A
Associative	(A·B)·C = A·(B·C)	(A+B)+C = A+(B+C)
Distributive	A·(B + C) = AB + AC	A + BC = (A + B)(A + C)
Absorption	A · (A + B) = A	A + A·B = A
Useful	A · (A′ + B) = AB	A + A′B = A + B
De Morgan's	**(A · B)′ = A′ + B′**	**(A + B)′ = A′ · B′**

Duality principle
Every Boolean identity stays true if we swap **AND ↔ OR** and **0 ↔ 1**. (That is why each law above has two forms.)

De Morgan's theorems in words

- The complement of a product = the sum of the complements. • The complement of a sum = the product of the complements.
- In circuits: NAND = OR of inverted inputs; NOR = AND of inverted inputs (bubbled gates).

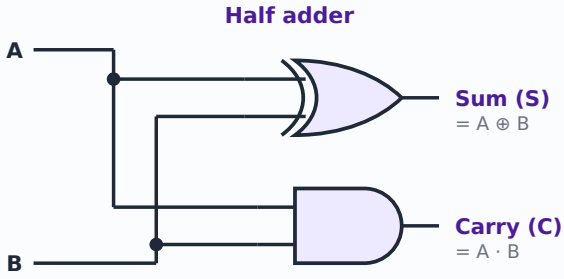
A	B	(A·B)′	A′ + B′	(A + B)′	A′ · B′
0	0	1	1	1	1
0	1	1	1	0	0
1	0	1	1	0	0
1	1	0	0	0	0

Simplification example
Simplify Y = A′B + AB′ + AB.
Y = A′B + A(B′ + B) = A′B + A·1 = A′B + A = **A + B** (using A + A′B = A + B). So the three-term circuit is just one OR gate.

5 Simple Combinational Circuits: Half and Full Adder

DEFINITION
A **combinational circuit** has outputs that depend **only on the present inputs** (no memory). Examples: adders, subtractors, multiplexers, decoders, encoders.

Half adder: adds two 1-bit numbers



A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Fig 3.4 Half adder: logic diagram and truth table (Sum = A ⊕ B, Carry = A · B)

Full adder: adds two bits and a carry-in

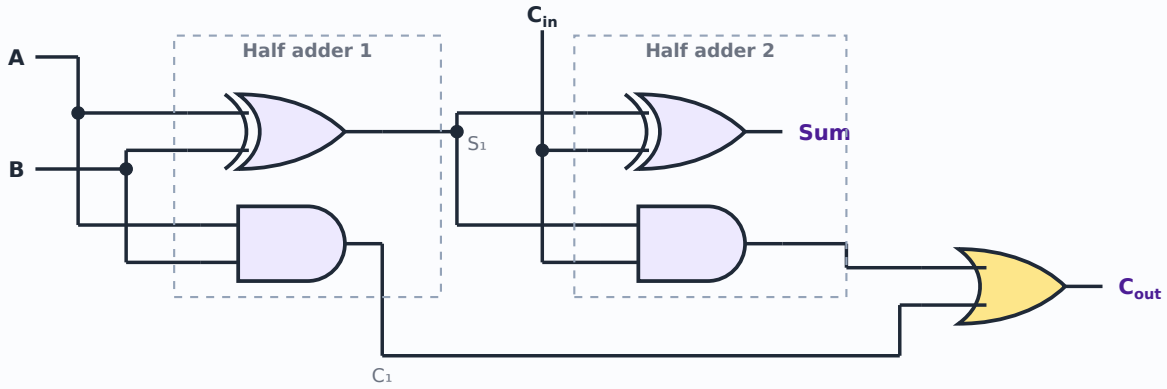


Fig 3.5 Full adder built from two half adders and an OR gate

A	B	C_{in}	Sum	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Adder equations

Half adder: $S = A \oplus B$, $C = A \cdot B$

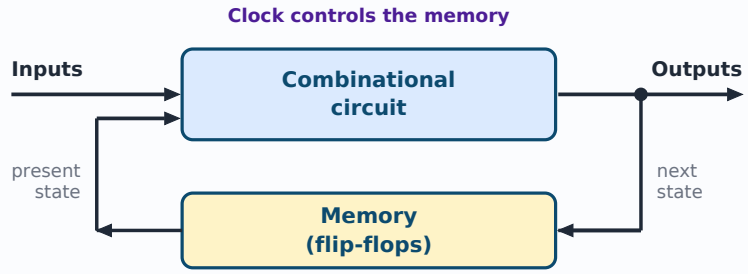
Full adder: $Sum = A \oplus B \oplus C_{in}$, $C_{out} = A{\cdot}B + C_{in}{\cdot}(A \oplus B)$

APPLICATIONS

Arithmetic Logic Unit (ALU) of every processor, calculators, address calculation, counters and digital signal processing. Cascading full adders gives an *n*-bit parallel adder.

6 Introduction to Sequential Circuits and Flip-Flops

	Combinational circuit	Sequential circuit
Output depends on	Present inputs only	Present inputs **and past state**
Memory element	None	Flip-flops / latches
Clock	Not needed	Usually clocked
Examples	Adder, decoder, multiplexer	Registers, counters, memories

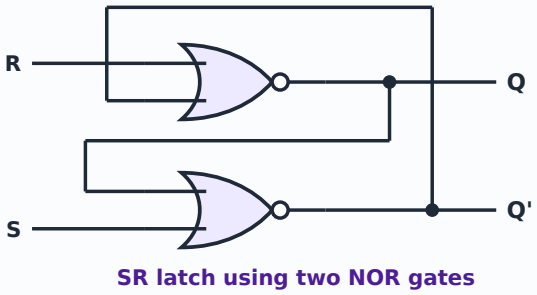


Flip-flop = a 1-bit memory cell with two stable states (Q = 0 or Q = 1). It remembers its state until the inputs and clock change it.

Latch responds to inputs at any time (level sensitive); a **flip-flop** changes only at the clock edge (edge triggered).

A flip-flop has outputs **Q** and its complement **Q′**.

Fig 3.6 Block diagram of a sequential circuit



Working of the NOR latch

S = 1, R = 0 → **Q = 1** (set)

S = 0, R = 1 → **Q = 0** (reset)

S = 0, R = 0 → **holds** the previous state (memory)

S = 1, R = 1 → **forbidden** (both outputs 0)

Fig 3.7 SR latch: the basic memory element

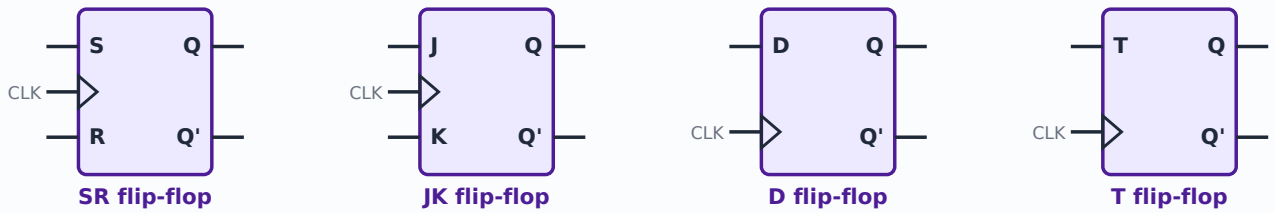


Fig 3.8 Block symbols of SR, JK, D and T flip-flops (triangle = clock input)

S	R	Q^+
0	0	Q
0	1	0
1	0	1
1	1	X

J	K	Q^+
0	0	Q
0	1	0
1	0	1
1	1	Q′

D	Q^+
0	0
1	1

T	Q^+
0	Q
1	Q′

Fig 3.9 Characteristic tables: SR, JK, D and T (Q^+ = next state; X = not allowed)

Flip-flop	Behaviour	Characteristic equation	Remark
SR	Set / Reset / Hold	$Q^+ = S + R'Q$	S = R = 1 is invalid
JK	Like SR, but 11 → **toggle**	$Q^+ = JQ' + K'Q$	Universal flip-flop; race-around in level-triggered form (cured by master-slave / edge triggering)
D	Output follows D after the clock (delay)	$Q^+ = D$	Used in registers and data storage
T	Toggles when T = 1	$Q^+ = T \oplus Q$	JK with J = K = T; used in counters

7 Registers and Counters (Elementary Treatment)

Register
A group of flip-flops that stores a binary word. An *n*-bit register uses *n* flip-flops. A **shift register** also moves the stored bits left or right on every clock pulse.

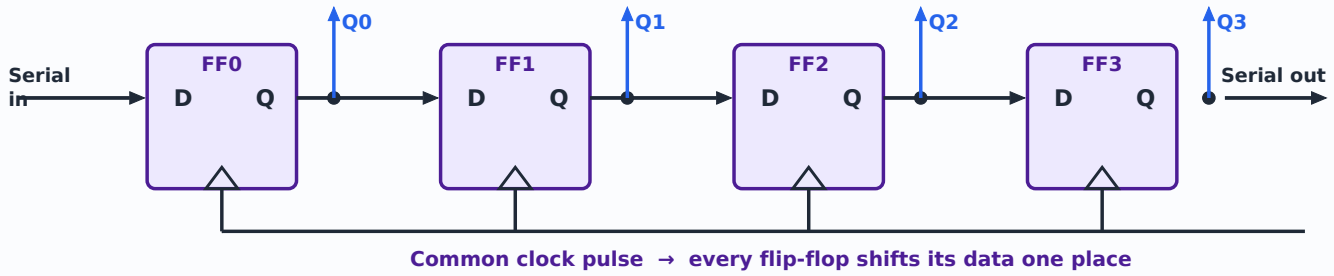


Fig 3.10 4-bit shift register (serial-in, parallel/serial-out) using D flip-flops

Type	Data in	Data out	Typical use
SISO	Serial	Serial	Time delay of digital signals
SIPO	Serial	Parallel	Serial-to-parallel conversion (receiver)
PISO	Parallel	Serial	Parallel-to-serial conversion (transmitter)
PIPO	Parallel	Parallel	Temporary data storage (buffer register)

Counters

DEFINITION
A **counter** is a sequential circuit that goes through a fixed sequence of states when clock pulses arrive. An *n*-bit counter has up to $\mathbf{2^n}$ states (a **mod-2^n** counter).

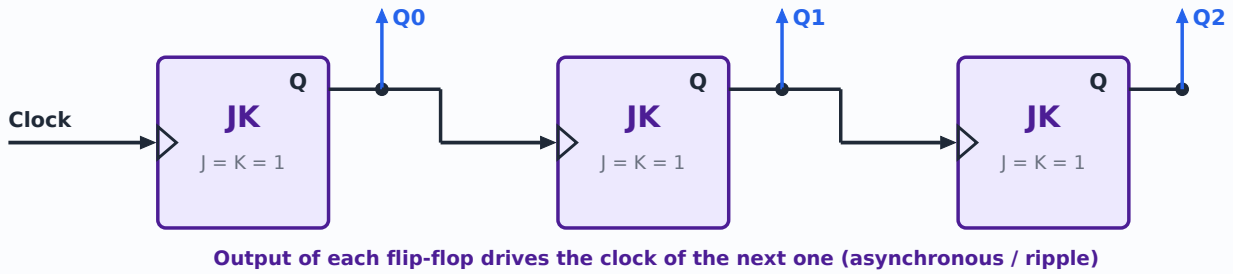
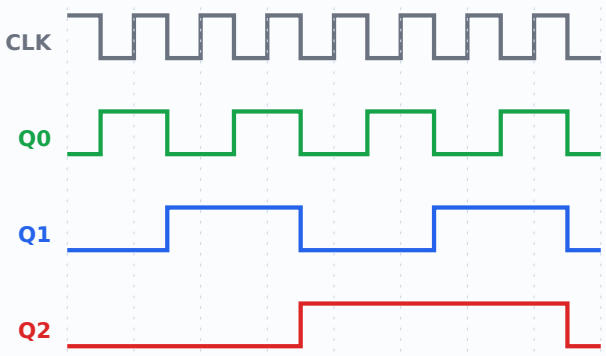


Fig 3.11 3-bit asynchronous (ripple) up-counter using JK flip-flops in toggle mode



Clock	Q2	Q1	Q0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0

Fig 3.12 Timing diagram and count sequence of the 3-bit counter (each stage divides the frequency by 2)

	Asynchronous (ripple) counter	Synchronous counter
Clock	Clock drives only the first FF; each FF clocks the next	**Same clock** to all flip-flops
Speed	Slower (delays add up)	Faster
Circuit	Simple, fewer gates	Needs extra gates; more complex

- **Up / down counters** count 0→7 or 7→0 (down counter takes the clock from Q′). **Mod-N counter:** resets after N states, e.g. a **decade (mod-10) counter** uses 4 flip-flops and clears at 1010.
- **Applications:** digital clocks and timers, frequency dividers, event counting, address generation in memories.

8 Quick Summary and Exam Corner

Unit III in one glance

- Binary (base 2), octal (8), hex (16). Group bits in 3s for octal and 4s for hex. 2’s complement = 1’s complement + 1.
- BCD = digit-by-digit 8421; Excess-3 = BCD + 3; Gray = one-bit change; Hamming (7,4) = 3 parity bits at 1, 2, 4 correct a single error.
- Gates: AND, OR, NOT, NAND, NOR, XOR, XNOR; NAND and NOR are universal. De Morgan: $(AB)' = A'+B'$, $(A+B)' = A'B'$.
- Half adder: $S = A \oplus B$, $C = AB$. Full adder: $S = A \oplus B \oplus Cin$, $Cout = AB + Cin(A \oplus B)$.
- Sequential = combinational + memory. Flip-flops: SR, JK (toggle at 11), D ($Q^+ = D$), T (toggle).
- Shift registers: SISO, SIPO, PISO, PIPO. n flip-flops give a mod-2^n counter; ripple = asynchronous, others = synchronous.

Short answer questions (2 marks)

1. Convert $(45)_{10}$ to binary, and $(1011\ 1100)_2$ to hexadecimal.
2. What is a universal gate? Name two.
3. State De Morgan’s theorems.
4. Why is Gray code used in position encoders?
5. Write the Sum and Carry expressions of a half adder.
6. Differentiate combinational and sequential circuits.
7. What is a flip-flop? How many states does it have?
8. What is the use of a shift register?

Long answer questions (5 / 10 marks)

1. Explain BCD, Excess-3 and Gray codes with a table for the digits 0 to 9.
2. Explain the Hamming code. Encode 1011 using even parity and show how a single-bit error is corrected.
3. Draw the symbol, truth table and expression of all basic gates (NOT, OR, AND, NOR, NAND, XOR, XNOR).

4. Show that NAND and NOR gates are universal by realising NOT, AND and OR gates using each.
5. State and prove the basic theorems of Boolean algebra, including De Morgan's theorems, using truth tables.
6. Design a half adder and a full adder; give truth tables and logic diagrams.
7. Explain the SR, JK, D and T flip-flops with symbols and characteristic tables.
8. Explain a 4-bit shift register and a 3-bit ripple counter with diagrams and timing.

Numerical practice (with answers)

1. Convert $(110101)_2$ to decimal, octal and hex. *Ans: 53, 65_8, 35_{16}*
2. Find the 2's complement of 01100. *Ans: 10011 + 1 = 10100*
3. Convert binary 1100 to Gray code. *Ans: 1010*
4. Simplify Y = AB + AB′. *Ans: Y = A(B + B′) = A*
5. How many flip-flops are needed for a mod-12 counter? *Ans: 4 (since $2^3 = 8 < 12 \le 2^4 = 16$)*

Quick check: pick the right answer

1. The output of a 2-input XOR gate is 1 when the inputs are: (a) same (b) different (c) both 1. *Ans: (b)*
2. Which flip-flop toggles for J = K = 1? (a) SR (b) D (c) JK. *Ans: (c)*
3. A 3-bit counter has how many states? (a) 3 (b) 6 (c) 8. *Ans: (c)*
4. Excess-3 code of decimal 4 is: (a) 0100 (b) 0111 (c) 1000. *Ans: (b)*